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Microstructures, magnetic properties and microwave absorption of ion-implanted bismuth ferrite thin films

Wiwien Andriyanti [a,b], Maureen Annisatul Choir Hidayati Nur [a], Deska Lismawenning Puspitarum [a,c], Tjipto Sujitno [d], Hari Suprihatin [e], Setyo Purwanto [b], Edi Suharyadi [a,*]

[a] Department of Physics, Universitas Gadjah Mada, Yogyakarta, Indonesia
[b] Research Center for Advanced Material, National Research and Innovation Agency, Indonesia
[c] Department of Physics, Institut Teknologi Sumatera, Lampung, Indonesia
[d] Research Center for Accelerator Technology, National Research and Innovation Agency, Indonesia
[e] Directorate of Laboratory Management, Research Facilities, and Science and Technology Park, National Research and Innovation Agency, Indonesia

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ABSTRACT

This study investigates the properties and performance of bismuth ferrite thin films on Kapton sheets as electromagnetic (EM) wave-absorbing materials. The main techniques employed in this study involved modifying the surface geometry and using electromagnetic wave-absorbing materials as surface coatings. X-ray diffraction analysis revealed a cubic crystal structure of the bismuth ferrite thin films. The bonding analysis showed the presence of Bi–O and Fe–O functional groups at 432.26 and 519.48 cm^{-1}, respectively. Vibrating-sample magnetometry revealed the ferromagnetic behavior of the bismuth ferrite thin film, confirming that saturation magnetization decreased with increasing sputtering time. Moreover, the calcination temperature affected the coercive field; the higher the calcination temperature, the higher the coercivity of the $BiFeO_3$ (BFO) thin film layer. Scanning electron microscopy with energy-dispersive X-ray spectroscopy of the bismuth ferrite thin films revealed the presence of Bi, Fe, and O—evenly and homogeneously distributed on the Kapton surface. After the ion implantation process, the atomic force microscopy test results showed an increased root-mean-square roughness (R_q). The root-mean-square roughness value for Kapton before the process was approximately 7.5 nm, while the root-mean-square roughness values for BFO 10 min and BFO 12.5 min were approximately 18.8 nm and 25.3 nm, respectively. The microwave absorption ability reached a maximum reflection loss value of −40.4 dB (99.9 %) in the range of 9.98 GHz with a sputtering duration of 10 min. Therefore, these results suggest bismuth ferrite is a potential candidate as an EM wave absorber.

1. Introduction

In recent years, research and development related to electromagnetic (EM) wave-absorbing materials has garnered interest because of their potential applications in various fields. These absorber materials convert EM wave energy into heat, making them ideal for protecting objects from unwanted EM waves. The main requirements for EM wave-absorbing materials are high permeability (magnetic loss properties) and permittivity (dielectric loss properties) [1–3].

Materials extensively studied for EM-wave absorption include $MnFe_2O_3$ [2,4], $LaMnO_3$ [5], $NiFeO_3$ [6], and $BiFeO_3$ (BFO) [7–14]. BFO is a rare single-phase multiferroic material with ferroelectric and antiferromagnetic properties at relatively high Curie and Néel temperatures of 810 °C and 375 °C, respectively. At room temperature (RT), BFO shows considerable potential for practical application owing to its large ferroelectric polarization (~110 μC/cm^2) and its distinctive behaviors by electric field [15–17], thus highlighting its potential for magnetoelectric (ME) applications at RT [18–20]. Multiferroics are materials that exhibit multiple magnetic, electrical, and elastic properties simultaneously. These materials are ME composites and produce the ME effect as a result of the mechanical coupling between piezoelectric (ferroelectric) and magnetostrictive (ferromagnetic) materials [21,22]. The single phases of these materials have several drawbacks in exhibiting the ME effect at room temperature, i.e., only observable in the

* Corresponding author.
E-mail address: esuharyadi@ugm.ac.id (E. Suharyadi).

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composite structure and absent in the constituent phases. A high ME effect can be attained by using proper piezomagnetic and piezoelectric phases, combining ferroelectric and ferri/ferromagnetic materials in an appropriate ratio, adding dopant atoms to multiferroic materials, incorporating various ions for Fe^{3+}, and controlling the preparation methods, including the sintering temperature and time [23–27]. The magnetic properties of multiferroic materials depend on the type of cations and their substitution distribution among the tetrahedral and octahedral positions, as well as the preparation method [28,29].

Single-phase bismuth ferrite (BFO) obtained from this synthesis is promising for future material applications [30]. However, the use of BFO presents several challenges, including the synthesis of multiferroic BFO with minimal impurities in the form of single-phase nanoparticles [31]. Achieving single-phase synthesis of multiferroic BFO can be challenging because of the increasing presence of impurities in the samples. Therefore, developing an optimal thin-film fabrication process for BFO is crucial for minimizing contamination-related problems.

BFO thin films can be fabricated using various methods, including physical techniques such as sputtering [32], pulsed laser deposition [33], and molecular beam epitaxy [34], and chemical methods, such as metal-organic chemical vapor deposition [35], sol-gel [36,37], and spray pyrolysis [38]. However, physical methods, such as sputtering, pulsed laser deposition, and molecular beam epitaxy, can produce single-phase BFO with minimal impurities. Among these techniques, DC sputtering has the advantages of easy production scalability, high deposition rates, and relatively low processing costs [39]. While BFO itself has some disadvantages, such as current leakage, weak magnetization properties, and poor electrical properties, the necessary improvements in its physical properties can be achieved through material modifications or engineering.

DC sputtering, a promising method for fabricating thin films on substrates, involves breaking the bonds between atoms in the target material, causing the atoms to rearrange on the substrate and form a new thin layer. In this study, thin films were fabricated on the surface of Kapton polymer film substrates. Kapton films are an exceptionally appealing group of polymer materials with unique properties that make them ideal for various industrial applications. Kapton has excellent thermal and chemical stability, low dielectric constant, strong electrical resistance, planar structure, reasonably easy transition into coating and forming processes, and high flexibility because it can be laminated, perforated, formed, or coated [40–44].

Previous studies [7] have reported the successful synthesis of multiferroic BFO nanoparticles using the sol-gel method, with the analysis results indicating that the BFO nanoparticles exhibited good absorption of electromagnetic waves. Upadhyay et al. [45] reported the successful growth of BFO thin film nanostructures via pulsed laser deposition (PLD). The resulting BFO films could be used in microwave applications. Kang et al. [46] reported that BFO nanoparticles were successfully synthesized by a sol-gel method and that the high reflection loss of BFO nanoparticles is favorable for their application as microwave absorption materials. According to previous research, a slight modification to the initial structure could change the magnetic and dielectric properties, eventually enhancing the microwave-absorbing properties. This study involved modifying the surface geometry and using electromagnetic wave-absorbing materials as surface coatings to improve the microwave-absorbing properties, which, to the best of our knowledge, has never been reported.

A novel aspect of this study is the creation of a nanogranular thin-film structure using a noble gas ion implantation process, such as argon, to enhance the properties of thin films. Modifications to the surface geometry to engineer the scattering characteristics include a nanogranular structure. As an EM wave-absorbing material, BFO offers several advantages in its nanogranular form. The nanogranular structure increased the surface area owing to the small grain size of the nanocrystals, allowing for more interactions in the absorption of EM wave energy within the absorber material. Another benefit is that its thickness can be adjusted to satisfy the physical parameter specifications required to achieve absorber characteristics with appropriate impedance [30,47]. Ion implantation is widely recognized as one of the most effective procedures for modifying the surface properties of polyimides. The change in surface properties due to ion implantation is reportedly caused by the difference in the structure and shape of the materials applied. The advantages of ion implantation techniques include the following: the process is fast and performable at room temperature, no dimensional changes occur, and the implantation depth and atom distribution can be controlled [48–50].

Considering the above observations, the present study aims to investigate the properties and performance of BFO thin films on Kapton sheets as electromagnetic (EM) wave-absorbing materials, with the morphological, magnetic, and microwave-absorbing properties of BFO thin films evaluated in a comprehensive study. As a result, this study necessitates investigating the fabrication and engineering of granular BFO thin films modified by argon ion implantation on Kapton/polyimide substrates to create composite films that are lightweight, flexible, and exhibit improved EM wave absorption.

2. Materials and methods

2.1. Preparation of the thin films

The materials used in this study were Kapton polyimide ($C_{22}H_{10}N_2O_5$) as the substrate, a BFO target, distilled water, and ethanol. The Kapton substrate—ultrasonically cleaned in distilled water and ethanol—was cut into 4 × 2 cm specimens. BFO thin films were deposited on Kapton substrates by DC sputtering under argon gas. The source voltage and current were maintained at 2.7 kV and 10 mA, respectively. The deposition process time parameters were varied from 5, 7.5, and 10–12.5 min. The detailed deposition conditions for the BFO thin films are listed in Table 1.

After the thin layers were formed, argon-ion implantation was performed to modify the surfaces of the BFO films. Argon-ion implantation was performed using a 150 keV/2 mA ion implanter at the Yogyakarta Radiation Laboratory, BRIN. Ion implantation of the samples was performed at a dose of 1×10^{16} ions/cm^2 with 45 keV argon ions. A schematic illustration of the BFO thin film deposition procedure is shown in Fig. 1.

The next step was the calcination of the BFO films. The calcination process involved placing the thin-film BFO samples into a furnace: one sample at 200 °C and another at 400 °C. The samples were maintained at these temperatures for 1 h after reaching the target temperature, with a calcination temperature rate of 10 °C per min.

2.2. Characterization

The surface morphologies of the BFO thin films were characterized using an Aeris benchtop X-ray diffractometer (XRD) to verify the phases. XRD verifications were performed on samples using radiation with a wavelength of 1.54 Å by keeping the scanning rate at 3.16 deg/min in voltage and current of 40 kV and 15 mA, respectively. Fourier transform infrared (FTIR) spectra were recorded using an Alpha II Bruker spectrometer to determine the functional groups. The surface morphology

Table 1
Detailed deposition conditions of BFO thin films issued from the DC sputtering.

Deposition	Target	BFO 99.99 % – diameter 1 in (Able Target Limited Company)
	Substrate	Kapton sheet 130 μm
	Voltage	2.7 kV
	Current	10 mA
	Target-substrate distance	2.5 cm

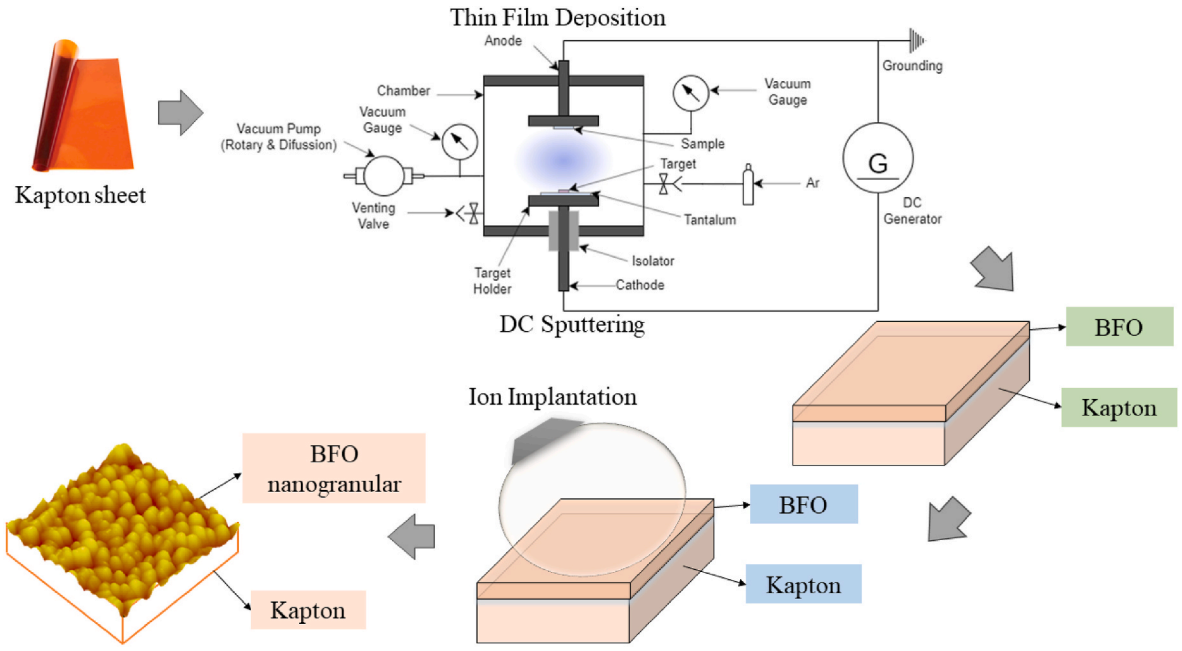


Fig. 1. Schematic illustration of BFO thin film deposition procedure.

and chemical composition of the BFO were determined using scanning electron microscopy (SEM) with energy-dispersive X-ray spectroscopy (Hitachi SEM-EDX SU3500). Three-dimensional topographic information on the sample surface was obtained using a commercial Park NX10 atomic force microscope (AFM). The magnetic behavior and EM absorption properties were analyzed using a Dexing vibrating sample magnetometer (model VSM 250) and a vector network analyzer (Anritsu VNA MS46322A).

3. Results and discussion

3.1. Crystalline structure and compositions

Fig. 2 shows the diffraction pattern of the BFO thin film deposited on the Kapton surface by DC sputtering. Based on the diffraction pattern results, the sharpness of the peaks increased with the duration of sputtering. The main peak at a 2θ angle around 43° corresponds to the (200) plane peak of bismuth ferrite ($Bi_{25}FeO_{40}$). The phase identification refers to COD 00-403-0661 for bismuth ferrite ($Bi_{25}FeO_{40}$) and COD 00-101-1164 for iron oxide, which has a cubic crystal structure.

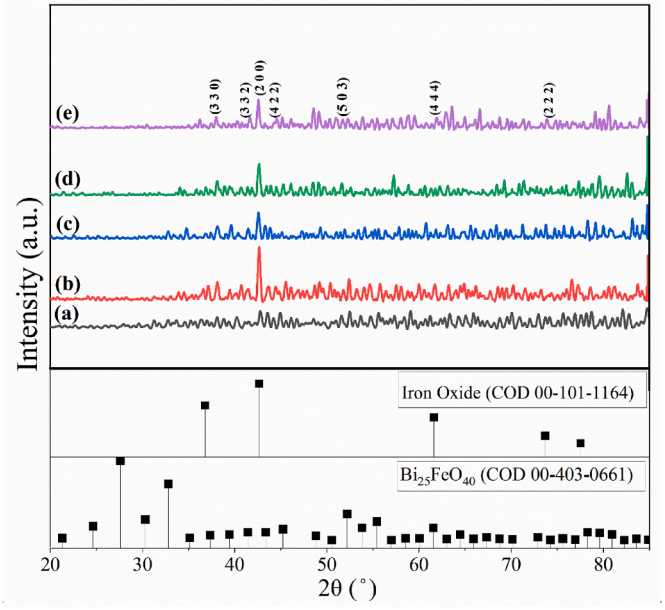


Fig. 2. XRD patterns of (a) Kapton (b) BFO 5 min, (c) BFO 7.5 min, (d) BFO 10 min, and (e) BFO 12.5 min.

The diffraction pattern graph reveals that the (200) plane peak becomes sharper and taller as the duration of the sputtering process increases, indicating that with longer sputtering time, the degree of crystallinity of the BFO thin film increases, leading to the deposition of more atoms onto the Kapton surface and the subsequent formation of a more significant number of crystals. During X-ray irradiation, X-rays scatter after being emitted from a crystal, with the scattering direction depending on the crystal structure. A consistent and regular crystal structure backscatters X-rays in the same direction, resulting in the constructive interference of the scattered waves and producing higher-intensity diffraction peaks [51].

The crystal size and lattice parameters of each time-varying sample were determined from the diffraction peak data, as shown in Table 2. Stabilizing the lattice positions of BFO atoms during sputtering is challenging. The 10-min sputtering sample had the largest crystal size of 55 nm among the time-variation samples. A larger crystal size occurs when crystal growth is more dominant, with fewer nuclei [51].

Table 2 also shows that there were no changes in the lattice parameter values among the time-varying samples. The lattice parameters of the samples refer to the constant distance between neighboring unit cells within the crystal lattice [52]. The consistent lattice parameter values across all the time-varying samples indicated that the interatomic distance was precise for each sample. These results indicated that the time variation in the sputtering process did not significantly affect the interatomic distances in the BFO thin films.

Table 2
Crystallite size and lattice parameter of BFO with various deposition times.

Sample	Crystallite Size (nm)	Lattice Parameter (Å)
BFO 5 min	45	8.0
BFO 7.5 min	41	8.0
BFO 10 min	55	8.3
BFO 12.5 min	39	8.0

Fig. 3 shows the diffraction pattern of the BFO thin films subjected to calcination at temperatures of 200 °C and 400 °C. The diffraction peaks corresponding to the diffraction planes in Fig. 2 were indexed to the (330), (332), (200), (422), (503), (444), and (222) planes. Table 3 lists the crystallite sizes and lattice parameters of the BFO thin films, indicating that the crystallite size increased at higher calcination temperatures. The heat generated during calcination increases the atomic mobility on the surface of the substrate, which tends to nucleate in its lowest energy state and forms larger structures by agglomerating around the nucleation sites [53], further supported by the tendency of the lattice parameters to decrease with increasing calcination temperature as the agglomerated atoms form larger structures, reducing the inter-unit cell distances within the crystallites.

3.2. Bonding analysis

FTIR spectroscopy is an essential tool for the bonding analysis of samples. The FTIR spectra of Kapton, BFO 5 min, BFO 7.5 min, BFO 10 min, and BFO 12.5 min were recorded in the wavenumber range of 400–4000 cm^{-1} and are presented in Fig. 4.

As seen in Fig. 4, the Kapton absorption peaks dominate the visible spectra. Vibrations corresponding to O–H in the wavelength range of 3200–3400 cm^{-1} and C=O in the wavelength range of 2100–2250 cm^{-1}, originating from Kapton [54], are evident. Additionally, the absorption peaks attributed to the vibrations of the Fe–O bonds at a wavelength of 519.48 cm^{-1} and Bi–O bonds at a wavelength of 432.26 cm^{-1} [55–59] can be observed. The presence of Fe–O and Bi–O bonds and the shift indicate the formation of BFO on the Kapton substrate. A shift in the absorption peaks of each bond in each sample was observed upon closer examination of the specific spectrum peaks. The constituent atoms of BFO, namely Bi, Fe, and O, can bind to atoms with different atomic numbers; therefore, the increase in wavenumber suggests the possibility of competition between the Fe or Bi atoms from BFO and the O atoms from Kapton [60]. An enlarged view of the FTIR spectra of the Fe–O and Bi–O functional groups is shown in Fig. 5.

Fig. 6 shows the FTIR spectra of Kapton with BFO deposited at different calcination times. Differences in the functional groups resulting from variations in the calcination time were mainly observed in the O–H bonds, with their peaks gradually decreasing as the calcination temperature increased, attributed to the evaporation of oxygen and hydrogen during calcination. The bonds of BFO did not change significantly during calcination.

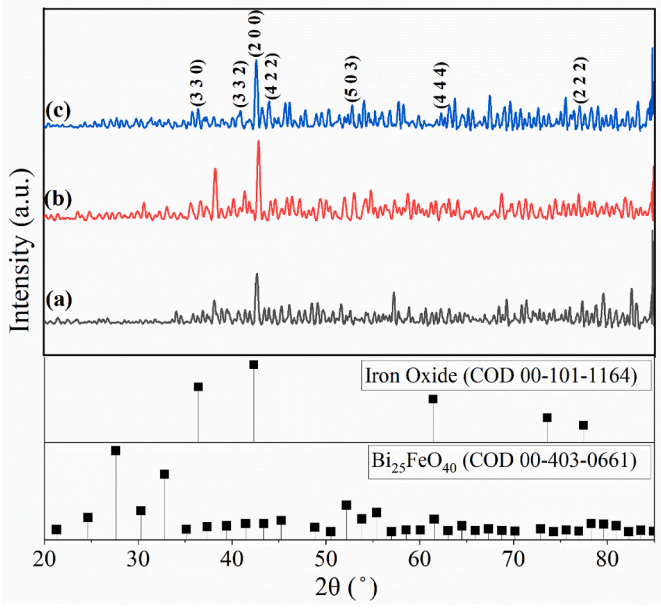


Fig. 3. XRD patterns of (a) non calcination and calcination at (b) 200 °C and (c) 400 °C.

Table 3
Crystallite size and lattice parameter of BFO with various calcination temperature.

Sample	Crystallite Size (nm)	Lattice Parameter (Å)
BFO non calcination	55	8.3
BFO 200 °C	202.1	8.2
BFO 400 °C	325	6.8

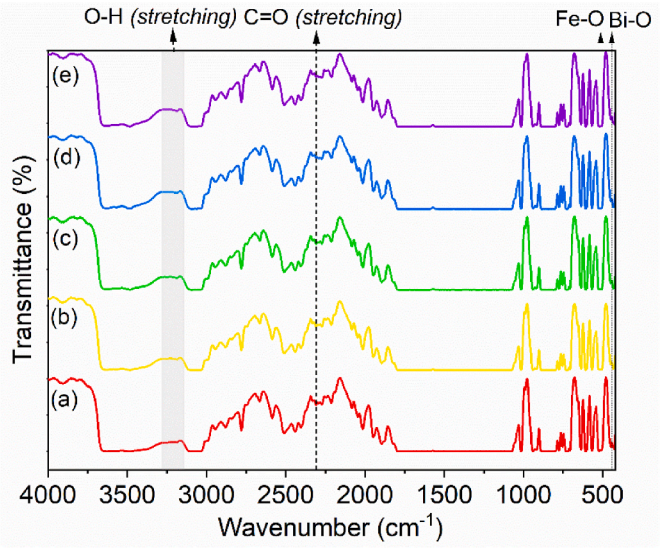


Fig. 4. FTIR spectra of (a) Kapton, (b) BFO 5 min, (c) BFO 7.5 min, (d) BFO 10 min, and (e) BFO 12.5 min.

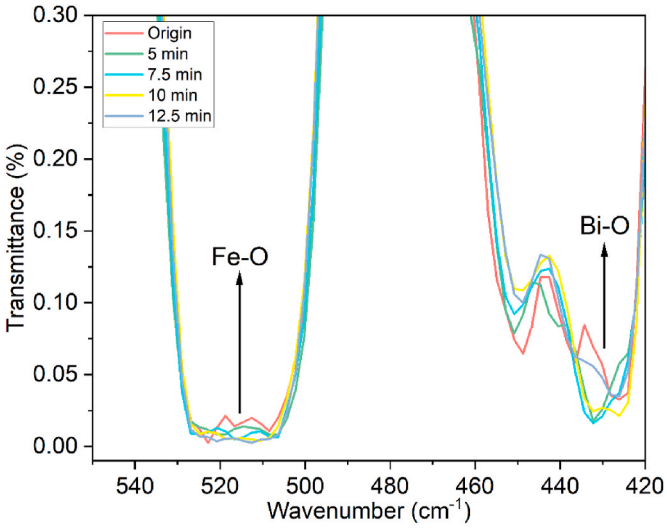


Fig. 5. Enlargement view of the FTIR spectrum of Fe–O and Bi–O functional groups.

3.3. Morphology analysis

The results of the observations of the surface morphology of the BFO samples at a sputtering time of 12.5 min are shown in Fig. 7. The elemental composition of each sample was determined by energy-dispersive EDX. Fig. 7(a) shows the presence of Bi, Fe, and O— evenly and homogenously distributed on the Kapton surface. The Bi content

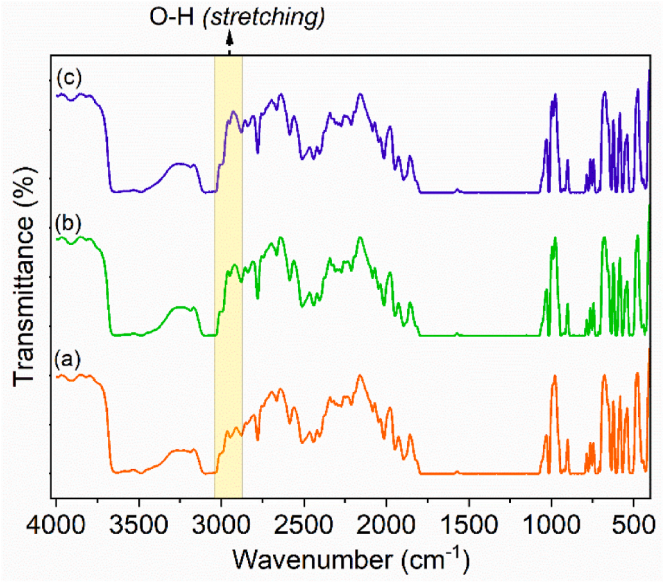


Fig. 6. FTIR spectra of (a) non calcination and calcination at (b) 200 °C and (c) 400 °C.

was 15 %, the Fe content was 18 %, and the O content was 67 % in weight percentages (wt. %). The presence of Bi, Fe, and O indicated the formation of a BFO thin film on the Kapton surface due to the DC sputtering deposition process. The surface morphology is continuous and homogeneous.

Cross-sectional SEM was conducted to determine the thickness of the thin films produced by the sputtering process. The thickness of the BFO thin film on the Kapton substrate was approximately 110 μm with a sputtering process time of 12.5 min. The obtained spectra indicate peaks corresponding to the Bi, Fe, and O elements with a thickness of 110 μm, suggesting successful deposition of BFO on the Kapton substrate. In this study, the thicknesses of the BFO thin films were determined using the deposition rate of DC sputtering. The deposition rate was obtained using SEM measurements of the thin-layer surface cross-section at a deposition period of 12.5 min. From the obtained thickness of the thin layer, the thicknesses of the thin layers at other deposition times were calculated and listed in Table 4. The grain sizes were determined from the SEM analysis results using the ImageJ software. The average grain size for a BFO of 12.5 min was 9.84 μm.

Fig. 8 presents the 2D and 3D AFM views of the Kapton origin and BFO thin films. In AFM, R_q is acquired from the root mean square (RMS), a function that considers the square of the measures that depend on the swept area of the sample. Before the ion implantation process, the surface was relatively smooth despite the presence of relatively small protrusions. The ion implantation led to a significant change in the surface morphology, with more significant areas of protrusions and valleys formed on the Kapton surface. Furthermore, the surface structure consisted of spherical grains, suggesting that the energy transfer during high-fluence implantation is sufficiently high to trigger the formation of such structures [49].

The RMS roughness values of the Kapton origin and BFO at various deposition times are listed in Table 5. The AFM test results showed an increased roughness after ion implantation. The root-mean-square roughness value for Kapton before the process was approximately 7.5 nm, while the root-mean-square roughness values for BFO 10 min and BFO 12.5 min were approximately 18.8 nm and 25.3 nm, respectively. The observed granular shape is one of the surface geometry modifications for engineering scattering properties. In its granular form, BFO exhibits various advantages as an absorber material for EM waves. The nanogranular structure enhanced the surface area, allowing for more interactions in absorbing EM wave energy in the absorber material [30, 47].

3.4. Magnetic properties

The M − H curve in Fig. 9 shows that the hysteresis pattern of the BFO thin-film samples generally exhibits ferromagnetic behavior. The calculated magnetic parameters of the samples are listed in Table 6.

The M_s and M_r of the BFO thin films decreased as the sputtering time decreased because of the increasing number of Bi and O atoms sputtered onto the Kapton surface. Bi is known to exhibit diamagnetic properties; the magnetic moment per unit area decreases when Bi atoms are sputtered onto a Kapton surface. According to fundamental studies [37], thin films with particles smaller than 62 nm exhibit a significant increase in ferromagnetic response. The increased performance of the BFO thin films can be described as follows: the particle size remained within the range of 62 nm, leading to the modulation of spiral spin moments, resulting in a greater enhancement of the ferromagnetic properties [37, 61]. With even smaller particle sizes, the BFO material demonstrated a negligible ferromagnetic response owing to the insufficient compensation of the spin moments. M_r of a material indicates the remaining magnetization after the removal of the external magnetic field, suggesting strong magnetic material characteristics.

The H_c values of the samples with different sputtering times varied irregularly. These inconsistent changes in the coercivity field can be associated with variations in the crystal size. In a magnetic material, when the particle size is proportional to the crystal size, boundaries between single-domain and multi-domain magnets appear at the critical particle sizes. Single-domain magnetic boundaries result in higher coercivity values when the particle size increases, whereas multi-domain magnetic boundaries can lead to lower coercivity values when the particle size decreases [62], implying that a larger magnetic field was required to change the magnetic spin orientation in the single-domain region as the crystal size decreased. Single-domain and multi-domain schematics and their influences on the coercivity field are illustrated in Fig. 10.

Furthermore, other factors such as the Bohr magneton, magnetic domain, crystallite size, and grain size are considered to play roles in determining the differences in coercivity [63]. In addition to the grain size, defects such as oxygen vacancies and Bi vacancies can influence domain switching and, consequently, the coercive field. These defects can interfere with the capacity of the domains to shift their orientation in response to an external input (such as an electric or magnetic field). Consequently, domain switching can affect the coercive field, which is the intensity of the external field required to induce polarization or magnetization reversal, possibly attributed to the larger grain size, which requires a larger electric field to switch domains [61].

Fig. 11 displays the M − H curve for the samples subjected to calcination at 200 °C and 400 °C. The M_s, or the maximum magnetization, of the thin-film BFO samples decreased as the calcination temperature increased. Mr represents the remaining magnetization after removing the external magnetic field, while M_r represents the remaining magnetization. Table 6 shows that the M_r values do not change significantly with increasing calcination temperature. The H_c values of the samples calcined at different temperatures increased with the calcination temperature.

The magnetic parameters of the BFO thin films calcined at different temperatures are listed in Table 7. The H_c value of the material increased with the calcination temperature owing to the increase in the thermal energy of the material, which could disrupt the alignment of the magnetic spins. Consequently, a single-domain material at room temperature splits into smaller and irregularly oriented domains, called multiple domains, at higher temperatures. At the boundary of the multi-domain magnetism, the material coercivity decreased as the particle size decreased [64]. Crystal-size calculations validated this statement. At room temperature, the crystallite size of the sample was 34.6 nm with an H_c of 766.6 Oe. At 200 °C, the crystallite size of the sample increased to

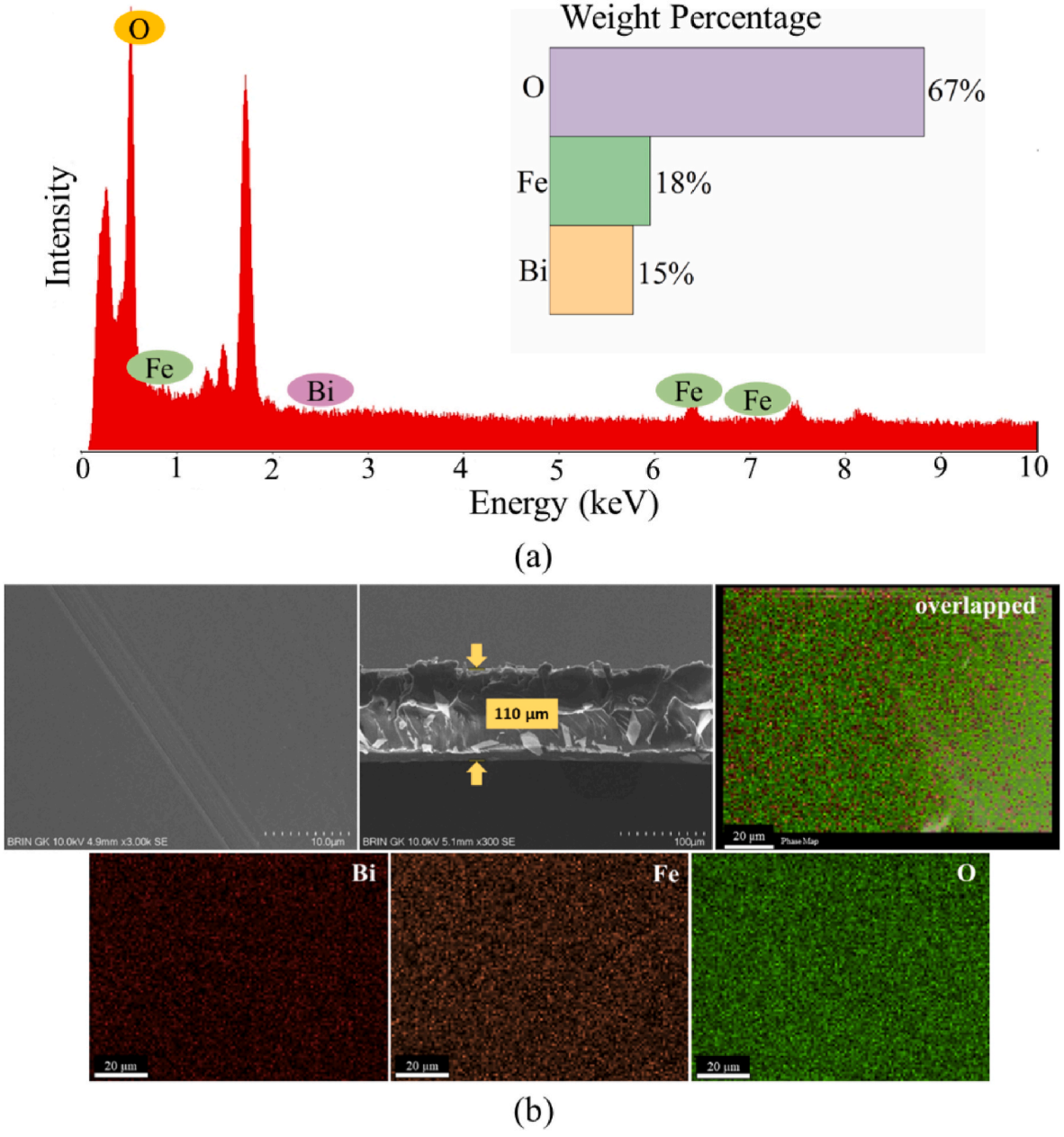


Fig. 7. (a) EDX spectrum, and (b) mapping analysis of BFO 12.5 min.

Table 4
Thin film thickness values with various deposition times.

Sample	Thickness (µm)
BFO 5 min	44
BFO 7.5 min	66
BFO 10 min	88
BFO 12.5 min	110

41 nm, accompanied by an increase in H_c to 927.6 Oe; this increase continued with the calcination of the sample: at 400 °C, the crystallite size was 55 nm and H_c was 1067.9 Oe. The table shows that the remanent magnetization values of each sample did not change significantly, leading to the assumption that the calcination temperature did not affect the remanent magnetization of the thin-film BFO layer.

3.5. EM absorbent analysis

According to microwave absorption theory, the incident power can be divided into three parts when EM waves strike the absorbing materials: reflection, absorption, and transmission. The transmitted wave is reflected on the surface of the conductive layer, causing many internal reflections. The reflection of EM waves comprises surface reflectance and multiple reflections [50]. A schematic representation of the EM absorber mechanism is shown in Fig. 12.

When EM waves interact with materials, the incident power can be divided into three components: reflection, absorption, and transmission. Incident EM energy can generate heat in the material through the interaction of the EM field with the molecular and electronic structures of the material, which can convert the incident EM waves into thermal energy, resulting in energy consumption. EM wave reflection consists of surface reflections and multiple reflections. Multiple reflections are

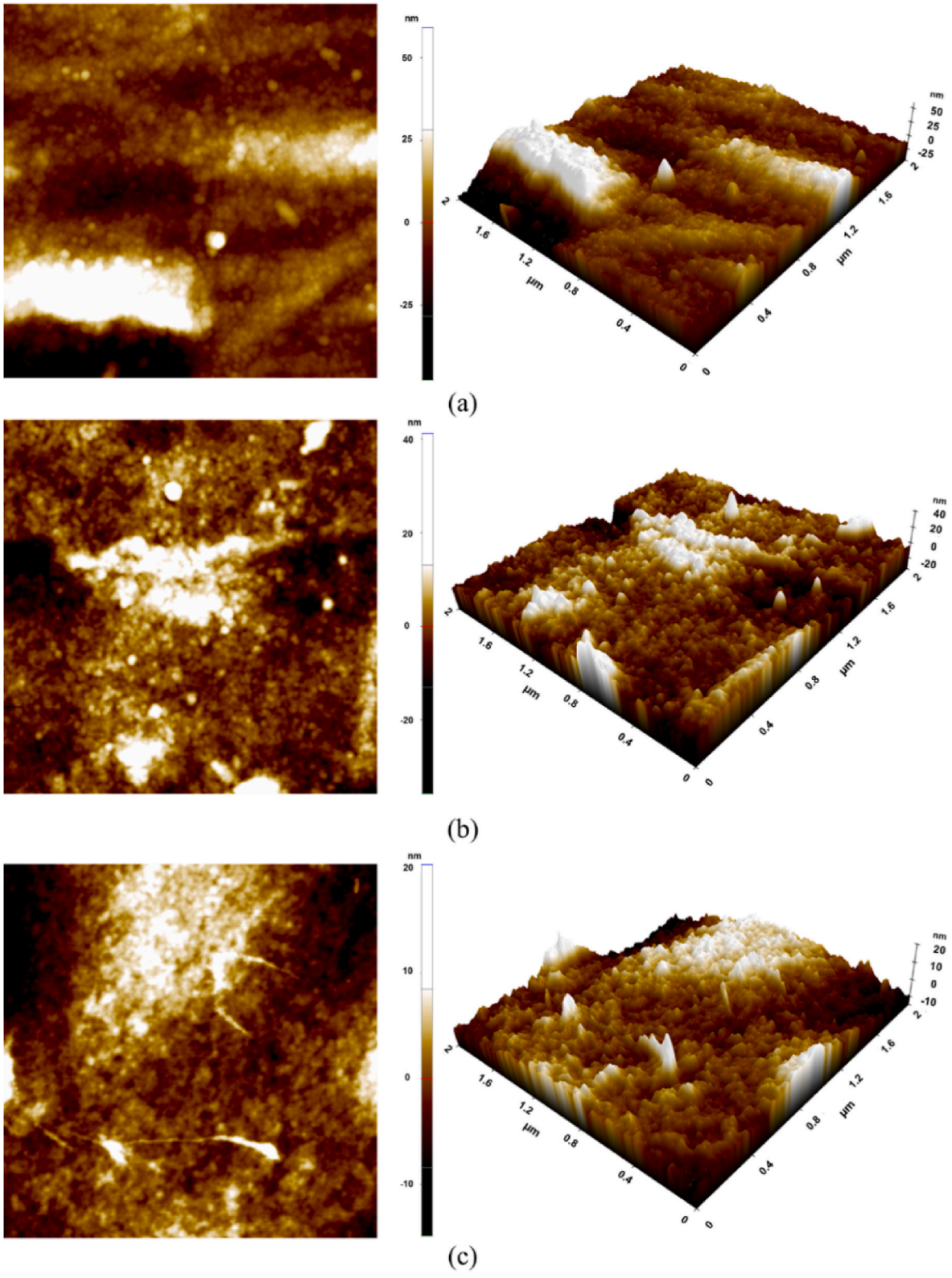


Fig. 8. AFM 2D and 3D image of (a) Kapton, (b) BFO 10 min, and (c) BFO 12.5 min.

generally considered a scattering effect of inhomogeneity within a material. Multiple reflections realized by adjusting the material structures (e.g., porous, hollow, core-shell, multilayer, and hierarchical) cause the propagation path of EM waves to extend and further improve the absorption capacity of the EM wave absorber [65,66]. The main techniques used in this study were surface geometry modification and the use of EM-wave-absorbing materials as surface coatings. The surface geometry of the nanogranular structure was designed to control scattering properties. By contrast, surface-layer materials absorb EM waves by converting their energy into another form [48]. Thus, more EM waves are absorbed by the absorber.

To assess the microwave absorption capabilities of each sample,

Table 5
RMS roughness values of Kapton origin and BFO with various deposition times.

Sample	RMS roughness value (nm)
Kapton origin	7.5
BFO 10 min	18.8
BFO 12.5 min	25.3

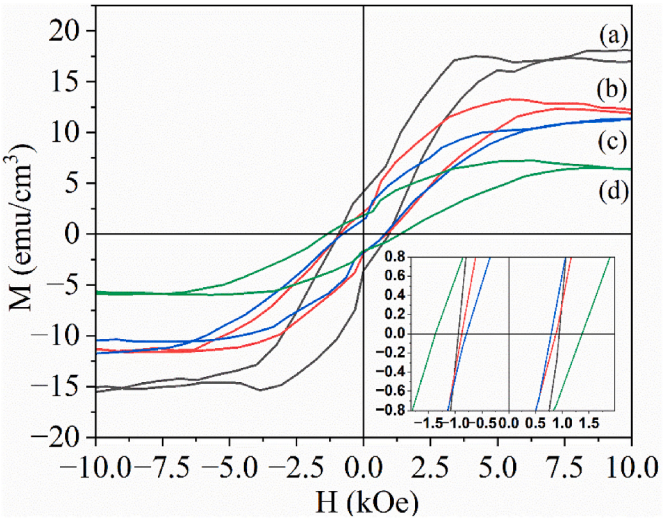


Fig. 9. *M-H* curve of (a) BFO 5 min, (b) BFO 7.5 min, (c) BFO 10 min, and (d) BFO 12.5 min.

Table 6
Saturation magnetization (M_s), coercivity (H_c), and remanent magnetization (M_r) of BFO with various deposition times.

Sample	M_s (emu/cm^3)	H_c (Oe)	M_r (emu/cm^3)
BFO 5 min	17.4	777.5	3.9
BFO 7.5 min	13.3	804.1	2.8
BFO 10 min	11.2	766.6	2.3
BFO 12.5 min	7.2	1283.6	2.2

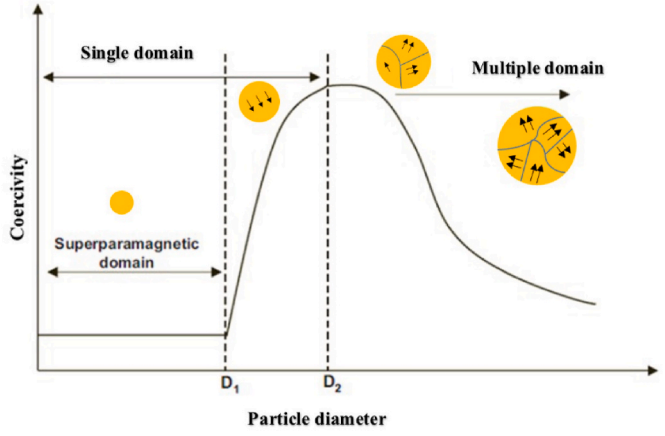


Fig. 10. Scheme of coercivity changes to magnetic particle size.

reflection loss (R_L) measurements were conducted using a VNA device (Anritsu VNA MS46322A) with a frequency range of 300 kHz to 20 GHz. According to the transmission line theory, the microwave absorption properties of materials can be determined by calculating their complex permeability and permittivity [30,37]. The loss of EM radiation

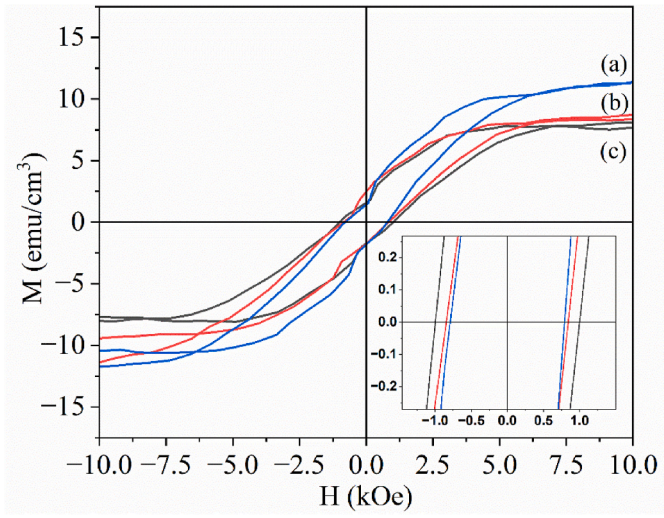


Fig. 11. *M-H* curve of (a) BFO non calcination and calcination at (b) 200 °C and (c) 400 °C.

Table 7
Saturation magnetization (M_s), coercivity (H_c), and remanent magnetization (M_r) of BFO with various calcination temperature.

Sample	M_s (emu/cm^3)	H_c (Oe)	M_r (emu/cm^3)
BFO non calcination	12.6	766.6	2.3
BFO 200 °C	8.7	927.6	1.8
BFO 400 °C	7.7	1067.9	1.9

reflected from a single-layer material can be calculated using the following equation [67]:

$$R_L = -20\,log\frac{(Z_{in} - Z_0)}{(Z_{in} + Z_0)} \quad (1)$$

$$\Gamma = 10^{\frac{-R_L}{20}} \quad (2)$$

$$Through\ Power\ (\%) = 100\left(1 - \Gamma^2\right) \quad (3)$$

The reflection loss phenomenon demonstrates the magnetic spin resonance mechanism between the initial electromagnetic wave or microwave and the magnetic spin of the material [68]. Fig. 13 shows the R_L values for the BFO samples at various deposition times within the 8–12 GHz frequency range. From these measurements, the S parameters for reflection (S11 and S22) and transmission (S21 and S22) were obtained. Using these parameters, the amount of reflection loss (RL) was calculated using Equation (1). For ease of understanding, the obtained RL values are often converted into a percentage of the microwave absorption attenuation, based on Equation (3). In the field application of a microwave absorber, an RL value of − 10 dB equals 90 % attenuation of the incident microwave energy. Thus, materials with such characteristics can be considered efficient microwave absorber candidates [69].

The calculations presented in Table 8 indicate a decreasing trend in R_L values for each process variation, signifying greater microwave absorption. The optimal R_L value was obtained for the BFO 10-min sample with a thin BFO layer and surface modification through ion implantation at 45 keV. The BFO 10-min sample exhibited the best performance as an EM wave-absorbing material, with R_L values of − 27.7 dB, 40.4 dB, and − 26.8 dB at frequencies of 8.76 GHz, 9.98 GHz, and 10.96 GHz, respectively. The effect of the deposition time also lowered and broadened the bandwidth, especially for BFO samples deposited for 10 and 12.5 min. The increased reflection losses were proportional to the material's coercivity (H_c); the larger the value of H_c, the greater the microwave absorption [63,70]. Modifying BFO thin films at various

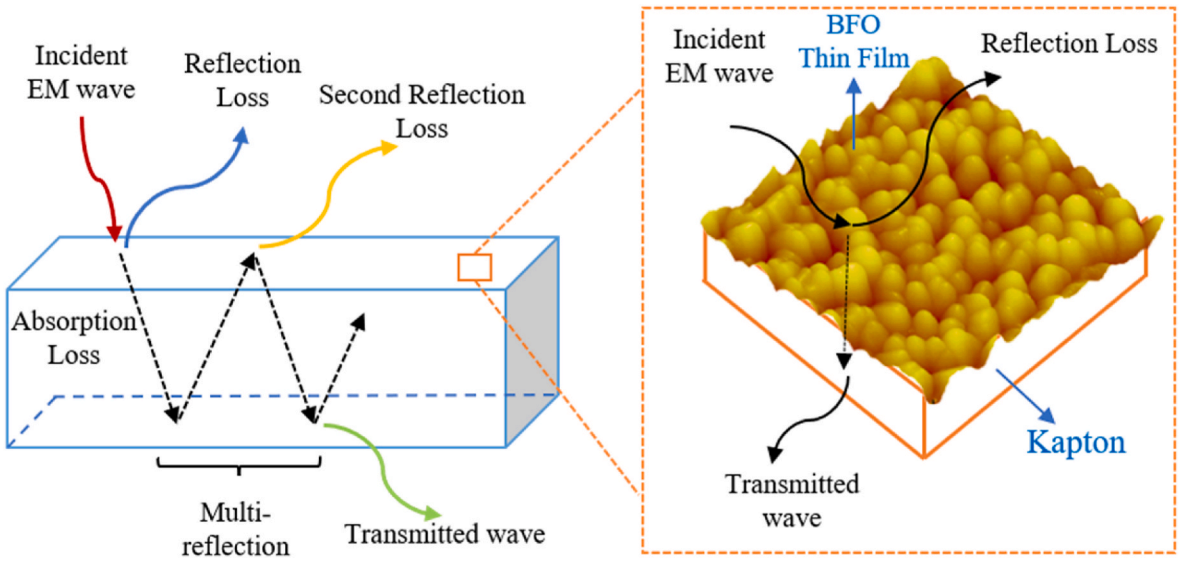


Fig. 12. Schematic representation of the EM absorber mechanism.

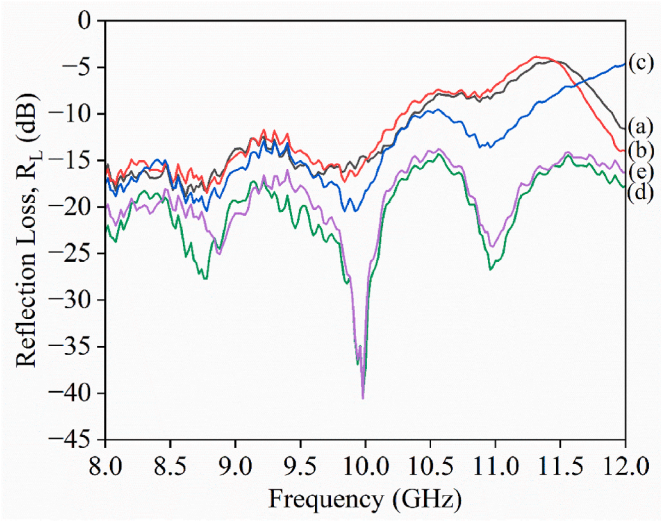


Fig. 13. Frequency dependence of Reflection Loss in (a) Kapton, (b) BFO 5 min, (c) BFO 7.5 min, (d) BFO 10 min, and (e) BFO 12.5 min.

Table 8
EM wave absorption of BFO implanted at 45 keV of ion energy for various deposition times.

Sample	Frequency (GHz)	Reflection Loss (dB)	Absorption wave (%)
Kapton origin	9.92	− 15.9	97.4
BFO 5 min	9.92	− 16.7	97.9
BFO 7.5 min	9.92	− 20.5	99.1
BFO 10 min	9.98	− 40.4	99.9
BFO 12.5 min	9.98	− 40.6	99.9

deposition times showed good results as EM-absorbing materials owing to their large coercivity values (Table 6).

The modifications applied to the Kapton material coated with BFO aimed to enhance the impedance matching between the free space (Z_0) and the material (Z_{in}). An increase in the Z_{in} value leads to a larger reflection loss (R_L) because the engineered surface geometry— designed to manipulate wave scattering—optimized the absorption of EM waves [47,70], consistent with the AFM data obtained, as displayed in Table 4, which shows increased roughness after the ion implantation process. The increase in surface roughness indicates the formation of nanogranules on the BFO thin-film layer. The nanogranular shape of this thin layer maximized the absorption of EM waves via a multiple-reflection mechanism. Multiple reflections can be controlled by altering the material properties (such as porosity, hollow structures, core-shell coatings, multi-layers, and hierarchical structures). These alterations allow the EM waves to develop propagation paths and enhance their potential for EM wave absorption [67].

As discussed earlier, a novel aspect of this study involves creating a nanogranular thin film structure using a noble gas ion implantation process, such as argon implantation, to enhance the properties of the thin films. The nanogranular structure is one of the modifications of the surface geometry aimed at engineering its scattering characteristics. In its nanogranular form, BFO offers various advantages as a microwave absorber material. The nanogranular structure increased the surface area owing to the small grain size of the nanocrystals, allowing for more interactions in absorbing electromagnetic wave energy within the absorber material. In this study, the microwave absorption capacity exceeds 90 %, highlighting the potential of bismuth ferrite as an EM wave absorber.

4. Conclusion

Bismuth ferrite (BFO) thin films were successfully grown on Kapton substrates via DC sputtering, with ion implantation used to modify the surface geometry of the absorbing material. The morphology, magnetic properties, and microwave absorption properties of the BFO thin films were investigated, with the existence of BFO thin films confirmed by the phase composition determined by XRD, SEM-EDX, AFM, and the binding from FTIR spectroscopy. Vibrating-sample magnetometry revealed the ferromagnetic behavior of the bismuth ferrite thin film, confirming that the longer the sputtering duration, the lower the Ms value (Ms) of the bismuth ferrite thin films. Moreover, the calcination temperature affected the coercivity field; the higher the calcination temperature, the higher the coercivity field value of the BFO thin film layer. Modifying the surface geometry allows for more interactions for absorbing electromagnetic wave energy within the absorber material, as demonstrated by the microwave absorption ability of the absorber material obtained in this study, i.e., 99.9 % in the range of 9.98 GHz. These results highlight the potential of bismuth ferrite material as an EM wave absorber.

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CRediT authorship contribution statement

Wiwien Andriyanti: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Maureen Annisatul Choir Hidayati Nur:** Formal analysis. **Deska Lismawenning Puspitarum:** Formal analysis, Validation, Writing – review & editing. **Tjipto Sujitno:** Investigation, Validation, Writing – review & editing. **Hari Suprihatin:** Investigation. **Setyo Purwanto:** Supervision, Validation, Writing – review & editing. **Edi Suharyadi:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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